Problem 4

- For each exercise provide the analytic solution (if it is possible to find an analytic solution, and include the parameter ranges)
- Otherwise, state that the analytic solution is impossible and provide the numerical solution (the correct choice is part of the evaluation)
- Calculate the entropy of the following distributions:
  1. $p(n) \propto \exp(-\lambda n)\,, n = 0,1,2, \dots$
  2. $p(n) \propto \exp(-n^2)\,, n = 0,1,2, \dots$
  3. $p(n) \propto n^{-4}, n = 1,2,3, \dots$
  4. $p(n) \propto \alpha^n, n = 1, \dots, N$
  5. $p(n) \propto (1 + n^2)^{-k}, n = 0,1,2, \dots$ for $k = 1,2$

HW2

Part A- Question 4

1.First we have to normalize the distribution.

$$Z = \sum_{n=0}^{\infty} \exp(-n) = \frac{1}{1 - \frac{1}{e}}.$$

The result is concluded from simple geometric series sum equation. Substituting this as p(x) in entropy formula, we'll have:

$$E = -\sum_{n=0}^{\infty} \frac{\exp(-n)}{Z} \ln\left(\frac{\exp(-n)}{Z}\right) = -\sum_{n=0}^{\infty} \frac{\exp(-n)}{Z} \left[-n - \ln(Z)\right] = \frac{1}{Z} \sum_{n=0}^{\infty} \exp(-n)n + \ln(Z).$$

Now we have to compute sum of $n * \exp(-n)$ from 0 to ∞ We know sum of x^n from 0 to ∞, and it is clear that we can just derivate it to reach the sum of the $n * x^n$ from 0 to ∞. Therefore, it is obvious that:

$$\sum_{n=0}^{\infty} n * x^n = \frac{x}{(1-x)^2}.$$

By substituting $\exp(-1)$ instead of x, we'll have:

$$\sum_{n=0}^{\infty} \exp(-n)n = \frac{\exp(-1)}{(1-\exp(-1))^2}.$$

So by substituting this and Z in the expression that we found for E, we'll have:

$$E = \frac{1}{Z}\sum_{n=0}^{\infty} \exp(-n)n + \ln(Z) = (1-\frac{1}{e}) * \frac{\exp(-1)}{(1-\exp(-1))^2} + ln(\frac{1}{1-\frac{1}{e}}) = \frac{\exp(-1)}{(1-\exp(-1))} + ln(\frac{1}{1-\frac{1}{e}})$$

2. We aim to calculate the entropy E of the distribution below, where Z is a normalizing factor:

$$p(n) = \frac{\exp(-n^2)}{Z}$$

but since the sum $\sum_{n=0}^{\infty} \exp(-n^2)$ does not have a closed form, we have to use numerical solutions.
Also, substituting the $p(n)$ given above in the entropy function, we will reach the expression below: (the last equality comes from the fact that the sum of the normalized distribution will be equal to 1)

$$S = -\sum_{n=0}^{\infty} \frac{\exp(-n^2)}{Z} \ln\left(\frac{\exp(-n^2)}{Z}\right) = -\sum_{n=0}^{\infty} \frac{\exp(-n^2)}{Z} \left[-n^2 - \ln(Z)\right]$$

$$= \frac{1}{Z} \sum_{n=0}^{\infty} \exp(-n^2) n^2 + \ln(Z).$$

To calculate this, we have to use numerical solutions. In order to do this, we used python code below:

And the results are:

$$Z \approx 1.386319, \quad S \approx 0.645665$$

```python
import math
s = 0
for n in range(0,100000):
  s += math.exp(-n**2)
E = 0
for n in range(0,100000):

  E += (math.exp(-n**2)*n*n)

print(s)
print(math.log(s)+(1/s)*(E))
```

```
1.3863186024133263
0.6456653446139602
```

Also I tried for 100 times more steps, but difference was less than 0.001.

3. First we try to normalize this distribution to have sum of 1. The sum in the following is known as the Riemann zeta function $\zeta(4)$.

$$Z = \sum_{n=0}^{\infty} n^{-4} = \zeta(4) = \frac{\pi^4}{90}.$$

Substituting , in the entropy function, we get:

$$E = -\sum_{n=1}^{\infty} \frac{n^{-4}}{Z} \ln\left(\frac{n^{-4}}{Z}\right) = -\sum_{n=1}^{\infty} \frac{n^{-4}}{Z} \left[-4\ln(n) - \ln(Z)\right] = \frac{4}{\frac{\pi^4}{90}} \sum_{n=1}^{\infty} n^{-4} \ln(n) + \frac{1}{\frac{\pi^4}{90}} \ln(\frac{\pi^4}{90})\zeta(4)$$

The first sum $\sum_{n=1}^{\infty} n^{-4} \ln(n)$ does not have a closed-form solution and requires numerical evaluation. Using a truncation at some large (e.g. 100) $n_{\max}$, we can compute this sum.
Thus, the entropy E can be approximated numerically by evaluating the sums. The Python code for calculating this value is as follows:

```
[19]
import math

s = 0
z = (math.pi**4)/90
for i in range(0,10000):
  s+= (n**(-4))*math.log(n)

print(4*s/z+math.log(z))
```

```
0.07910987306733964
```

4. The probability mass function $p(n)$ is proportional to a^n. To convert this into a valid probability distribution, we need to normalize it:

$$Z = \sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$$

Hence, the normalized probability distribution is:

$$p(n) = \frac{a^n}{1-a}, \quad n = 0, 1, 2, \ldots$$

By substituting this in entropy function, we'll have:

$$E = -\sum_{n=0}^{\infty} \frac{a^n}{1-a} \log\left(\frac{a^n}{1-a}\right) = -\sum_{n=0}^{\infty} \frac{a^n}{1-a} \left(\log(a^n) - \log(1-a)\right) =$$

$$= -\sum_{n=0}^{\infty} \frac{a^n}{1-a} \left(n\log(a) - \log(1-a)\right) = -\sum_{n=0}^{\infty} \frac{a^n}{1-a} n\log(a) + \sum_{n=0}^{\infty} \frac{a^n}{1-a} \log(1-a)$$

It is obvious that the second term is equal to $log(1-a)$. The first term is equal to $alog(a)/(1-a)$ times the derivation of Z, so the result is equal to:

$$H(X) = -\frac{a}{(1-a)^3} \log(a) + \log(1-a)$$

5. To calculate the entropy, we first have to normalize the probability:

For $k = 1$ and $k = 2$, these sums do not have closed-form solutions, but are related to the Hurwitz zeta function $\zeta(k)$. Therefore, for numerical evaluation, we approximate these sums using truncation at large $N_{\max}$.

Thus, the normalized probability distribution is as follows:

$$p(n) = \frac{(1+n^2)^{-k}}{Z_k}, \quad k = 1, 2$$

We've calculated their approximate value with a python code (see below).

Substituting this probability in entropy function, we'll have:

$$E = -\sum_{n=0}^{\infty} \frac{(1+n^2)^{-k}}{Z_k} \log\left(\frac{(1+n^2)^{-k}}{Z_k}\right) = -\sum_{n=0}^{\infty} \frac{(1+n^2)^{-k}}{Z_k} \left(-k\log(1+n^2) - \log Z_k\right)$$

$$= k\sum_{n=0}^{\infty} \frac{(1+n^2)^{-k}}{Z_k} \log(1+n^2) + \log Z_k \sum_{n=0}^{\infty} \frac{(1+n^2)^{-k}}{Z_k}$$

The second sum is equal to 1, hence:

$$H(X) = k\sum_{n=0}^{\infty} \frac{(1+n^2)^{-k}}{Z_k} \log(1+n^2) + \log Z_k$$

We cannot compute this analytically because we do not have closed-form solution of the first sum above for general k. These sums are related to special functions like the Hurwitz zeta function, and even though such functions exist, they do not yield simple closed-form expressions for entropy. Therefore, numerical methods are required to estimate the entropy. The calculated entropies are:

```python
import math
E1 = 0
E2 = 0
s1 = 0
s2 = 0
for n in range(0,1000000):
  s1 += (1+n**2)**(-1)
  s2 += (1+n**2)**(-2)

for n in range(0,1000000):
  n1 = ((1 + n**2)**(-1)) / s1
  n2 = ((1 + n**2)**(-2)) / s2
  E1 += n1*math.log(n1)
  E2 += n2*math.log(n2)
print('Z1 =', s1)
print('Z2 =', s2)
print('Entropy for k = 1 is', -E1)
print('Entropy for k = 2 is', -E2)
```

```
Z1 = 2.076673047468177
Z2 = 1.306836975422631
Entropy for k = 1 is 1.7742596267503163
Entropy for k = 2 is 0.7009278176290031
```